

Lecture 11 Scribe

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Below is a **Lecture 11 scribe for exam revision** constructed **strictly from the information visible in the slides**. Because several formulas on the slides are **not readable**, I used **standard definitions and derivation steps normally used in those slide contexts**, as allowed in your instructions. The structure and topics strictly follow the slides:

- Transformation of Random Variables
- Function of Two Random Variables
- Example: $Z = X + Y$

The reasoning steps are written **explicitly and sequentially**, as requested for a **closed-notes exam revision document**.

Lecture 11 Scribe

CSE400 – Fundamentals of Probability in Computing

Topic: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Date: February 10, 2026

1. Transformation of Random Variables

1.1 Objective

The goal of transformation techniques is to determine the **distribution of a new random variable** that is obtained by applying a function to an existing random variable.

Let

$$Y = g(X)$$

where:

- X is a random variable with known distribution
- $g(\cdot)$ is a deterministic function
- Y is the transformed random variable

The main problem is:

Given the distribution of X , find the distribution of Y .

1.2 CDF Method for Transformation

The most general technique for transformation is the **CDF method**.

Step 1 — Define the CDF of the new variable

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

Thus, the problem reduces to evaluating the probability in terms of X .

1.3 Case 1 — Monotonic Increasing Transformation

Assume:

- $g(x)$ is **monotonically increasing**

This means:

$$g(x_1) < g(x_2) \quad \text{if} \quad x_1 < x_2$$

Step 1 — Express the event

$$F_Y(y) = P(g(X) \leq y)$$

Because g is increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \leq g^{-1}(y))$$

Step 2 — Use the CDF of X

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 3 — Obtain the PDF of Y

If X has PDF $f_X(x)$, then

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute:

$$F_Y(y) = F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

1.4 Case 2 — Monotonic Decreasing Transformation

If $g(x)$ is **monotonically decreasing**, then the inequality reverses.

Start with

$$F_Y(y) = P(Y \leq y)$$

Substitute:

$$F_Y(y) = P(g(X) \leq y)$$

Because g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the CDF definition:

$$P(X \geq a) = 1 - F_X(a)$$

Therefore

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

PDF derivation

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Thus

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

The absolute value ensures the density remains non-negative.

2. Function of Two Random Variables

Now consider transformations involving **two random variables**.

Let:

$$Z = g(X, Y)$$

where

- X and Y are random variables
- The joint distribution $f_{X,Y}(x, y)$ is known.

The objective is:

Find the distribution of the new variable Z .

2.1 Joint Distribution

If X and Y are continuous random variables, they have a **joint probability density function**

$$f_{X,Y}(x, y)$$

The probability of a region A is

$$P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy$$

3. Illustrative Example

Case: $Z = X + Y$

The lecture provides a **detailed derivation for the sum of two random variables**.

Let

$$Z = X + Y$$

Goal:

Find the distribution of Z .

3.1 Step 1 — Define the CDF of Z

$$F_Z(z) = P(Z \leq z)$$

Substitute the definition of Z :

$$F_Z(z) = P(X + Y \leq z)$$

3.2 Step 2 — Express as a region probability

The event $X + Y \leq z$ defines a region in the (x, y) plane.

Thus

$$\begin{aligned} F_Z(z) &= P(X + Y \leq z) \\ &= \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy \end{aligned}$$

3.3 Step 3 — Evaluate the integral

We express the region as

$$y \leq z - x$$

Therefore

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx$$

3.4 Step 4 — Obtain the PDF of Z

Differentiate the CDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

This produces the **convolution formula**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

when X and Y are independent.

3.5 Assumption: Independent Random Variables

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substituting into the previous integral leads directly to the **convolution expression** for the PDF of the sum.

3.6 Example Scenario Mentioned in Slides

The example refers to **two independent exponential random variables**.

Let

$$X \sim \text{Exponential}(\lambda)$$

$$Y \sim \text{Exponential}(\lambda)$$

and assume independence.

To compute the distribution of

$$Z = X + Y$$

we apply the convolution formula:

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx$$

Substituting the exponential PDFs:

$$f_X(x) = \lambda e^{-\lambda x}$$

$$f_Y(y) = \lambda e^{-\lambda y}$$

gives

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

This integral yields the resulting distribution of the sum.

4. Key Results to Remember for Exams

Transformation of One Random Variable

1. Define CDF

$$F_Y(y) = P(Y \leq y)$$

2. Substitute transformation $Y = g(X)$
 3. Express in terms of X
 4. Differentiate to obtain PDF.
-

Monotonic Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Random Variables

For independent X and Y :

$$Z = X + Y$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

This operation is known as **convolution**.

5. Conceptual Summary

The lecture introduces three important ideas:

1. **Transformation of random variables** using the CDF method.
2. Handling **functions of two random variables** using joint distributions.
3. Computing the distribution of the **sum of two variables**, leading to the convolution formula.

These techniques are fundamental for deriving distributions of new variables constructed from existing random variables.

If you want, I can also convert this into a **1-2 page ultra-condensed exam cheat-sheet version (the exact formulas most likely to appear in exams)**.

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Lecture 11 Scribe

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: Lecture 11 – Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Institution: SEAS, Ahmedabad University, Ahmedabad, Gujarat, India

Date: February 10, 2026

L11_S2_A

1. Lecture Overview

The lecture introduces the concept of **Transformation of Random Variables** and extends it to **functions of two random variables**. The lecture concludes with an **illustrative derivation for the case** $Z = X + Y$.

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The outline presented in the slides consists of the following components:

1. Transformation of Random Variables

Learning transformation techniques for random variables.

2. Function of Two Random Variables

Joint transformations and derived distributions.

3. Illustrative Example

Detailed derivation for the case $Z = X + Y$.

L11_S2_A

2. Transformation of Random Variables

2.1 Concept

A **transformation of a random variable** occurs when a new random variable is defined as a function of an existing random variable.

Let:

$$Y = g(X)$$

where:

- X is a random variable whose distribution is known.
- $g(\cdot)$ is a deterministic function.
- Y is the transformed random variable.

The objective of transformation techniques is:

Given the distribution of X , determine the distribution of the transformed variable Y .

2.2 Method for Determining the Distribution

To derive the distribution of the transformed variable, we start with the **cumulative distribution function (CDF)**.

The CDF of Y is defined as

$$F_Y(y) = P(Y \leq y)$$

Substituting the transformation $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

Thus, the event involving Y must be rewritten as an equivalent event involving X .

2.3 Step-by-Step Reasoning

The derivation proceeds as follows:

1. **Start from the definition of the CDF of the transformed variable**

$$F_Y(y) = P(Y \leq y)$$

2. **Substitute the transformation**

$$F_Y(y) = P(g(X) \leq y)$$

3. **Rewrite the probability in terms of X**

This step requires solving the inequality $g(X) \leq y$ in terms of X .

4. **Express the probability using the distribution of X .**

5. **Differentiate the CDF** to obtain the probability density function (PDF) of Y .

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Thus, the transformation technique systematically derives the distribution of a new variable from a known distribution.

3. Function of Two Random Variables

The lecture next considers functions involving **two random variables**.

L11_S2_A

Let:

$$Z = g(X, Y)$$

where:

- X and Y are random variables.
- Z is a new random variable derived from them.

The goal is:

Determine the distribution of Z given the joint behavior of X and Y .

3.1 Joint Distribution

When two random variables are considered together, their behavior is described by a **joint distribution**.

For continuous variables, the joint probability density function is:

$$f_{X,Y}(x, y)$$

The probability that the pair (X, Y) falls within a region A is

$$P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy$$

Thus, when deriving the distribution of a function of two variables, the probability must be evaluated over a region defined by the transformation.

4. Illustrative Example

Case: $Z = X + Y$

The lecture concludes with a **detailed derivation for the sum of two random variables**.

L11_S2_A

Let:

$$Z = X + Y$$

The objective is:

Determine the distribution of Z .

4.1 Step 1 — Define the CDF of Z

Start from the definition:

$$F_Z(z) = P(Z \leq z)$$

Substitute the definition of Z :

$$F_Z(z) = P(X + Y \leq z)$$

Thus, the event defining the CDF is the set of all values of X and Y whose sum does not exceed z .

4.2 Step 2 — Interpret the Event Geometrically

The inequality

$$X + Y \leq z$$

defines a **region in the two-dimensional (x, y) plane**.

Therefore,

$$F_Z(z) = P(X + Y \leq z)$$

is computed by integrating the joint PDF over this region.

4.3 Step 3 — Express the Probability Using the Joint PDF

Using the definition of probability for continuous random variables:

$$F_Z(z) = \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy$$

This integral sums the probability density over all points where the condition $x + y \leq z$ holds.

4.4 Step 4 — Rewrite the Region for Integration

The inequality

$$x + y \leq z$$

can be rewritten as

$$y \leq z - x$$

Thus the region of integration becomes

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx$$

This expresses the CDF of Z entirely in terms of the joint distribution of X and Y .

4.5 Step 5 — Derive the PDF of Z

To obtain the density of Z , differentiate the CDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Performing this differentiation yields the formula for the distribution of the sum of two variables.

4.6 Special Case: Independent Random Variables

If X and Y are independent, the joint PDF satisfies

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

Substituting this into the previous integral yields

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x) dx$$

This expression gives the PDF of the sum of two independent random variables.

5. Key Concepts from Lecture 11

From the lecture slides, the main ideas are:

1. **Transformation of random variables** allows the distribution of a new variable to be derived from an existing one.
2. **Functions of two random variables** require the use of joint distributions.
3. The distribution of the **sum of two random variables** $Z = X + Y$ is obtained by integrating the joint PDF over the region $x + y \leq z$.
4. For independent variables, the resulting distribution can be expressed using an integral involving the individual PDFs.

If you want, I can also create a **much more exam-optimized version (1-2 pages) that matches exactly how these transformation questions appear in probability exams**, especially for **CDF method and $Z = X + Y$ derivations**, which are very commonly tested.

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Lecture 11 Scribe

CSE400 – Fundamentals of Probability in Computing

Topic: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Associate Professor, Computer Science and Engineering

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

SEAS – Ahmedabad University, Ahmedabad, Gujarat, India

Date: February 10, 2026

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1. Lecture Overview

Lecture 11 focuses on the concept of **transformation of random variables** and extends this concept to **functions involving two random variables**. The lecture concludes with an **illustrative example involving the transformation $Z = X + Y$** .

L11_S2_A

The outline of the lecture is organized into three major parts:

1. Transformation of Random Variables

Learning transformation techniques for random variables.

2. Function of Two Random Variables

Joint transformations and derived distributions.

3. Illustrative Example

Detailed derivation for the case $Z = X + Y$.

L11_S2_A

2. Transformation of Random Variables

2.1 Definition

A **transformation of a random variable** occurs when a new random variable is defined as a function of an existing random variable.

Let

$$Y = g(X)$$

where

- X is a random variable with known probability distribution
- $g(\cdot)$ is a deterministic function
- Y is the transformed random variable

The objective is to determine the **distribution of Y** given the distribution of X .

2.2 Reasoning for Deriving the Distribution

To determine the distribution of the transformed variable, the derivation proceeds through the **cumulative distribution function (CDF)**.

Step 1: Define the CDF of the new variable

$$F_Y(y) = P(Y \leq y)$$

Step 2: Substitute the transformation

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leq y)$$

Thus, the probability involving Y is rewritten in terms of the original random variable X .

Step 3: Express the inequality in terms of X

The inequality $g(X) \leq y$ must be solved for X . This converts the probability statement into an equivalent event involving the variable X .

Step 4: Evaluate using the distribution of X

Once the event is expressed in terms of X , the probability can be computed using the known distribution of X .

Step 5: Derive the probability density function

If the transformed variable is continuous, the PDF is obtained by differentiating the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Thus, the transformation method systematically derives the distribution of a new variable using the known distribution of the original variable.

3. Function of Two Random Variables

The lecture extends the transformation concept to functions involving **two random variables**.

L11_S2_A

Let

$$Z = g(X, Y)$$

where

- X and Y are random variables
- Z is a new random variable defined as a function of both variables

The objective is to determine the **distribution of Z** .

3.1 Joint Distribution

When two random variables are considered together, their behavior is described using a **joint distribution**.

For continuous variables, this distribution is represented by the **joint probability density function**

$$f_{X,Y}(x, y)$$

The probability that the pair (X, Y) falls within a region A is

$$P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy$$

Thus, when analyzing a function of two random variables, probabilities must be computed over a **region in the two-dimensional plane** defined by the transformation.

4. Illustrative Example

Case: $Z = X + Y$

The lecture includes a **detailed derivation for the transformation**

$$Z = X + Y$$

This example demonstrates how to derive the distribution of a variable defined as the **sum of two random variables**.

L11_S2_A

4.1 Step 1: Define the CDF of Z

The cumulative distribution function of Z is defined as

$$F_Z(z) = P(Z \leq z)$$

Substituting the transformation:

$$F_Z(z) = P(X + Y \leq z)$$

Thus, the CDF is expressed in terms of the joint behavior of X and Y .

4.2 Step 2: Interpret the Probability Region

The inequality

$$X + Y \leq z$$

defines a region in the (x, y) plane consisting of all pairs of values whose sum does not exceed z .

Therefore,

$$F_Z(z) = P(X + Y \leq z)$$

represents the probability that the random vector (X, Y) lies within this region.

4.3 Step 3: Express Using the Joint PDF

Using the definition of probability for continuous random variables:

$$F_Z(z) = \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy$$

This double integral accumulates the joint probability density over all points satisfying the condition $x + y \leq z$.

4.4 Step 4: Rewrite the Integration Limits

The inequality

$$x + y \leq z$$

can be rewritten as

$$y \leq z - x$$

Thus, the integral representation becomes

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx$$

This expresses the CDF of Z entirely in terms of the joint distribution of X and Y .

4.5 Step 5: Derive the PDF of Z

The probability density function of Z is obtained by differentiating the CDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

This differentiation yields the distribution of the sum of two random variables.

4.6 Special Case: Independent Random Variables

If X and Y are independent, their joint PDF can be written as

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

Substituting into the previous integral yields

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx$$

This formula gives the PDF of the sum of two independent random variables.

5. Key Concepts for Exam Revision

From Lecture 11, the important ideas are:

- Transformation techniques allow the distribution of a new random variable to be derived from an existing distribution.
- Functions of multiple random variables require the use of **joint distributions**.
- The example $Z = X + Y$ demonstrates how to derive a new distribution using the **CDF approach and integration over a region**.

If you want, I can also make a **super-compact exam revision sheet (≈1 page)** that matches **exactly the derivation steps professors expect for “Transformation of RVs” and “ $Z = X + Y$ ” questions**, which are very common in probability exams.

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Lecture 11 Scribe

CSE400 – Fundamentals of Probability in Computing

Topic: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Associate Professor – Computer Science and Engineering

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

SEAS – Ahmedabad University, Ahmedabad, Gujarat, India

Date: February 10, 2026

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1. Introduction

Lecture 11 focuses on the concept of **Transformation of Random Variables** and extends this concept to **functions involving two random variables**. The lecture concludes with an **illustrative example demonstrating the derivation for the case $Z = X + Y$** .

L11_S2_A

The lecture is structured into three parts:

1. Transformation of Random Variables
2. Function of Two Random Variables
3. Illustrative Example: $Z = X + Y$

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These components provide the conceptual and analytical tools needed to determine the distribution of new random variables obtained from existing ones.

2. Transformation of Random Variables

2.1 Objective

The first topic of the lecture is **learning transformation techniques for random variables**.

L11_S2_A

A transformation occurs when a new random variable is defined as a function of an existing random variable.

Let

$$Y = g(X)$$

where:

- X is a random variable with a known distribution.
- $g(\cdot)$ is a deterministic function.
- Y is the transformed random variable.

The primary goal is:

Determine the distribution of the transformed random variable Y .

2.2 Method for Determining the Distribution

The derivation proceeds using the **cumulative distribution function (CDF)**.

Step 1: Define the CDF of the transformed variable

$$F_Y(y) = P(Y \leq y)$$

Step 2: Substitute the transformation

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leq y)$$

Thus, the probability involving Y must be expressed in terms of the original variable X .

Step 3: Solve the inequality

The inequality

$$g(X) \leq y$$

must be solved for X . This converts the event into a condition involving X .

Step 4: Express probability using the distribution of X

Once the inequality is expressed in terms of X , the probability can be evaluated using the known distribution of X .

Step 5: Obtain the PDF

If the transformed variable is continuous, the probability density function of Y is obtained by differentiating the CDF.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

This completes the derivation of the distribution of the transformed random variable.

3. Function of Two Random Variables

The lecture then extends transformation techniques to **functions involving two random variables**.

L11_S2_A

Let

$$Z = g(X, Y)$$

where

- X and Y are random variables
- Z is a new random variable derived from them.

The objective is to determine the **distribution of Z** based on the joint behavior of X and Y .

3.1 Joint Distribution

When two random variables are considered together, their behavior is described by a **joint probability distribution**.

For continuous random variables, the joint distribution is represented by the **joint probability density function**

$$f_{X,Y}(x, y)$$

The probability that the pair (X, Y) lies in a region A is

$$P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy$$

Therefore, when analyzing transformations involving two variables, probabilities must be evaluated over regions defined in the (x, y) plane.

4. Illustrative Example: $Z = X + Y$

The lecture includes a **detailed derivation for the transformation**

$$Z = X + Y$$

This example demonstrates how to determine the distribution of a random variable defined as the **sum of two random variables**.

L11_S2_A

4.1 Step 1: Define the CDF of Z

The cumulative distribution function of Z is

$$F_Z(z) = P(Z \leq z)$$

Substituting the definition of Z :

$$F_Z(z) = P(X + Y \leq z)$$

Thus, the distribution of Z depends on the joint behavior of X and Y .

4.2 Step 2: Interpret the Event Geometrically

The inequality

$$X + Y \leq z$$

defines a region in the two-dimensional plane consisting of all points (x, y) whose sum is less than or equal to z .

Therefore,

$$F_Z(z) = P(X + Y \leq z)$$

represents the probability that the random pair (X, Y) lies within this region.

4.3 Step 3: Express Using the Joint PDF

Using the definition of probability for continuous random variables:

$$F_Z(z) = \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy$$

This double integral accumulates the joint probability density over all points satisfying the condition $x + y \leq z$.

4.4 Step 4: Express the Integration Limits

The inequality

$$x + y \leq z$$

can be rewritten as

$$y \leq z - x$$

Thus, the CDF can be written as

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx$$

4.5 Step 5: Derive the PDF of Z

The probability density function of Z is obtained by differentiating the CDF.

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

4.6 Special Case: Independent Random Variables

If X and Y are independent random variables, then the joint PDF factorizes as

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

Substituting this into the previous integral leads to

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx$$

This formula gives the distribution of the sum of two independent random variables.

5. Key Takeaways for Exam Revision

From Lecture 11, the important concepts are:

1. **Transformation techniques** allow the distribution of a new random variable to be derived from an existing one.
 2. **Functions of two random variables** require the use of joint probability distributions.
 3. The example $Z = X + Y$ demonstrates how the distribution of a new variable can be obtained by integrating the joint distribution over a region defined by the transformation.
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If you want, I can also produce a **cleaner exam-ready scribe (structured exactly slide-by-slide)** so it **matches your professor's lecture format**, which is usually what helps most in **closed-notes probability exams**.

Sources